

EXPERIMENT 03 – DYNAMIC DROPDOWNS AND CHECKBOXES

1. Module Name and Experiment Title

Module Name: Module 1 – Automation with Selenium

Experiment Number: 03

Experiment Title: Dynamic Dropdowns and Checkboxes

Programming Language: Python

Website: Rahul Shetty Academy – Automation Practice

Website URL: <https://rahulshettyacademy.com/AutomationPractice/>

2. Problem Statement

To automate the selection of multiple checkboxes and handle a dynamic autocomplete dropdown using Selenium WebDriver with Python.

The experiment uses the Rahul Shetty Academy Automation Practice website. The automation script must select specific checkboxes, verify their selection states, enter a country name into the autocomplete field, select the desired country from the suggestions, and verify the selected value.

3. Objective

The main objectives of this experiment are:

1. To automate multiple checkbox elements using Selenium WebDriver.
 2. To identify web elements using appropriate Selenium locators.
 3. To select checkboxes and verify their selection states using `is_selected()`.
 4. To handle a dynamic autocomplete dropdown.
 5. To enter text into an autocomplete input field using `send_keys()`.
 6. To wait for dynamically displayed suggestions using explicit waits.
 7. To select a specific suggestion from the available options.
 8. To validate the selected country using assertions.
 9. To understand how Selenium WebDriver interacts with dynamic web elements.
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4. Tools, Software and Concepts Used

Programming Language: Python

Automation Framework: Selenium WebDriver

Browser: Google Chrome

IDE: Visual Studio Code

Website: Rahul Shetty Academy – Automation Practice

Python Libraries:

- selenium
- time

Selenium Modules:

- webdriver
- By
- WebDriverWait
- expected_conditions

Concepts Used:

- WebDriver initialization
 - Browser maximization
 - Element identification using XPath and ID
 - Checkbox selection
 - is_selected() method
 - Autocomplete dropdown handling
 - Explicit waits
 - send_keys() and click() methods
 - Assertions
 - Exception-safe browser closure using finally
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5. Source Code / Implementation Steps

Source Code

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

# Initialize Chrome WebDriver
driver = webdriver.Chrome()
driver.maximize_window()

wait = WebDriverWait(driver, 15)

try:
```

```

# Open the Automation Practice website
driver.get("https://rahulshettyacademy.com/AutomationPractice/")

time.sleep(3)

# -----
# Task 1: Handle Multiple Checkboxes
# -----

checkboxes = wait.until(
    EC.presence_of_all_elements_located(
        (By.XPATH, "//input[@type='checkbox']")
    )
)

print("Total checkboxes:", len(checkboxes))

# Locate Option 1 and Option 3
option1 = driver.find_element(
    By.XPATH, "//input[@value='option1']"
)

option3 = driver.find_element(
    By.XPATH, "//input[@value='option3']"
)

# Select Option 1
if not option1.is_selected():
    option1.click()

time.sleep(3)

# Select Option 3
if not option3.is_selected():
    option3.click()

time.sleep(3)

# Verify checkbox selection
assert option1.is_selected()
assert option3.is_selected()

print("Option 1 selected:", option1.is_selected())
print("Option 3 selected:", option3.is_selected())

# -----
# Task 2: Handle Autocomplete Dropdown
# -----

autocomplete = wait.until(
    EC.visibility_of_element_located(
        (By.ID, "autocomplete")
    )
)

# Enter country name
autocomplete.clear()
autocomplete.send_keys("Ind")

time.sleep(3)

```

```

# Locate autocomplete suggestions
suggestions = wait.until(
    EC.visibility_of_all_elements_located(
        (By.CSS_SELECTOR, ".ui-autocomplete li")
    )
)

print("Autocomplete suggestions:")

for suggestion in suggestions:
    print(suggestion.text)

# Select India from the suggestions
india_found = False

for suggestion in suggestions:
    if suggestion.text.strip().casefold() == "india":
        suggestion.click()
        india_found = True
        break

assert india_found, "India suggestion not found"

time.sleep(3)

# Verify the selected country
selected_country = autocomplete.get_attribute("value")

assert selected_country == "India"

print("Selected country:", selected_country)
print("All tasks completed successfully!")

finally:
    driver.quit()

```

Implementation Steps

Step 1: Import Required Libraries

Import Selenium WebDriver, the `By` class, `WebDriverWait`, `expected_conditions`, and the `time` module.

Step 2: Initialize the WebDriver

Create a Chrome WebDriver instance and maximize the browser window.

Step 3: Open the Website

Navigate to the Rahul Shetty Academy Automation Practice website using the `driver.get()` method.

Step 4: Locate the Checkboxes

Locate all checkbox elements using XPath and count the available checkboxes.

Step 5: Select Multiple Checkboxes

Locate Option 1 and Option 3 using XPath. Select both checkboxes if they are not already selected.

Step 6: Verify Checkbox Selection

Use the `is_selected()` method to verify that Option 1 and Option 3 are selected. Assertions ensure that both checkboxes are successfully selected.

Step 7: Locate the Autocomplete Field

Locate the autocomplete input field using its ID, `autocomplete`.

Step 8: Enter the Country Name

Clear the input field and enter `Ind` using the `send_keys()` method.

Step 9: Handle Dynamic Suggestions

Wait for the autocomplete suggestions to become visible. Retrieve the suggestions and display their text in the terminal.

Step 10: Select the Desired Country

Iterate through the suggestions and select the option whose text matches `India`.

Step 11: Verify the Selected Country

Retrieve the input field's value using `get_attribute("value")` and verify that it equals `India`.

Step 12: Close the Browser

Use a `finally` block to ensure that the browser is closed after execution.

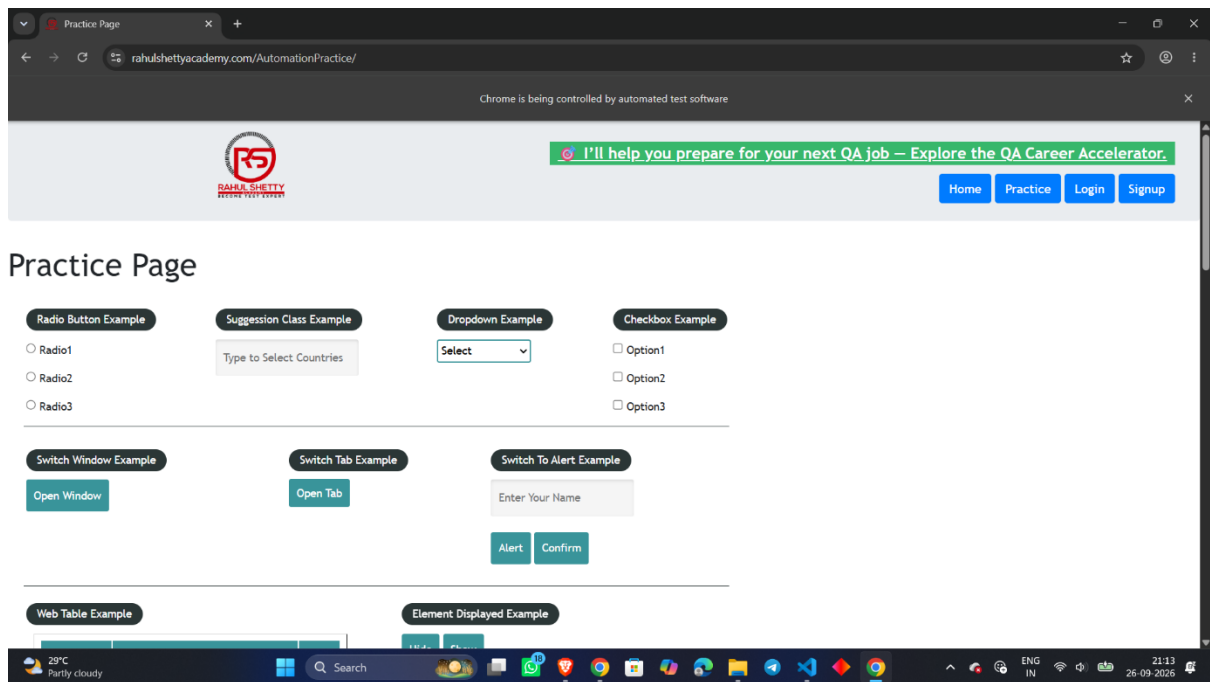
6. Output with Clearly Labelled Screenshots

The Selenium automation script was executed successfully on the Rahul Shetty Academy Automation Practice website.

The following screenshots demonstrate the initial state, the completed automation, and the console output.

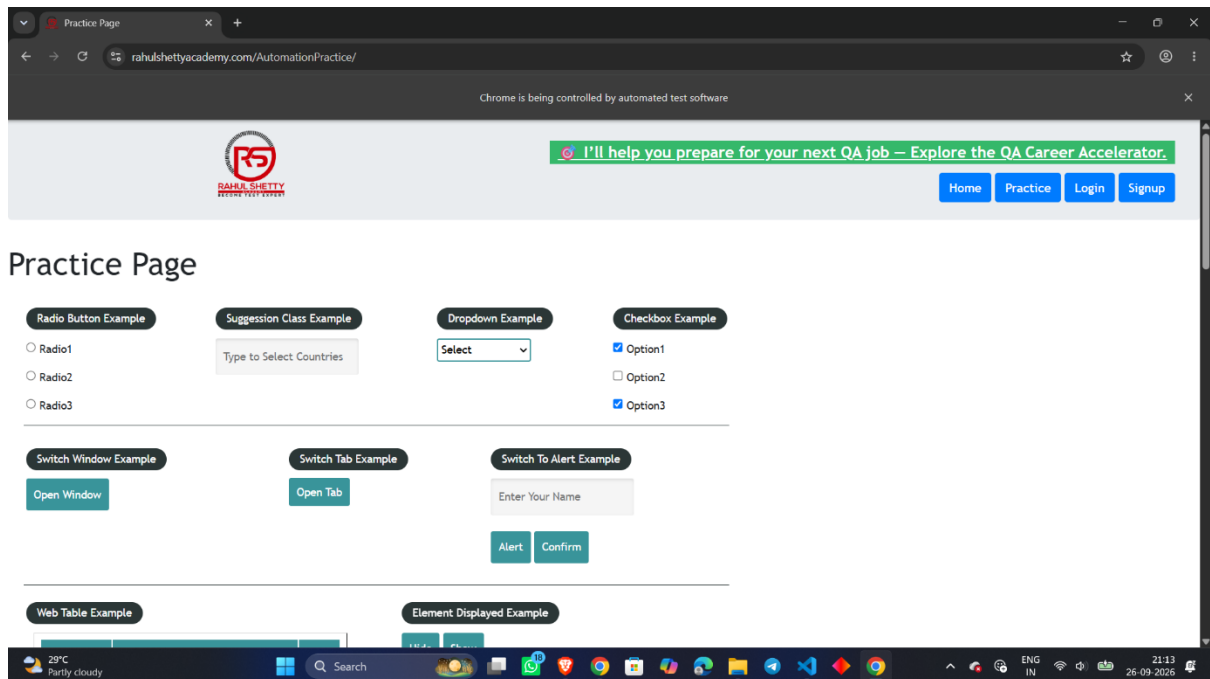
Screenshot 1: Browser in Initial State

Description: The Automation Practice website is displayed before the automation begins. All three checkboxes are unchecked, and the autocomplete field is empty.



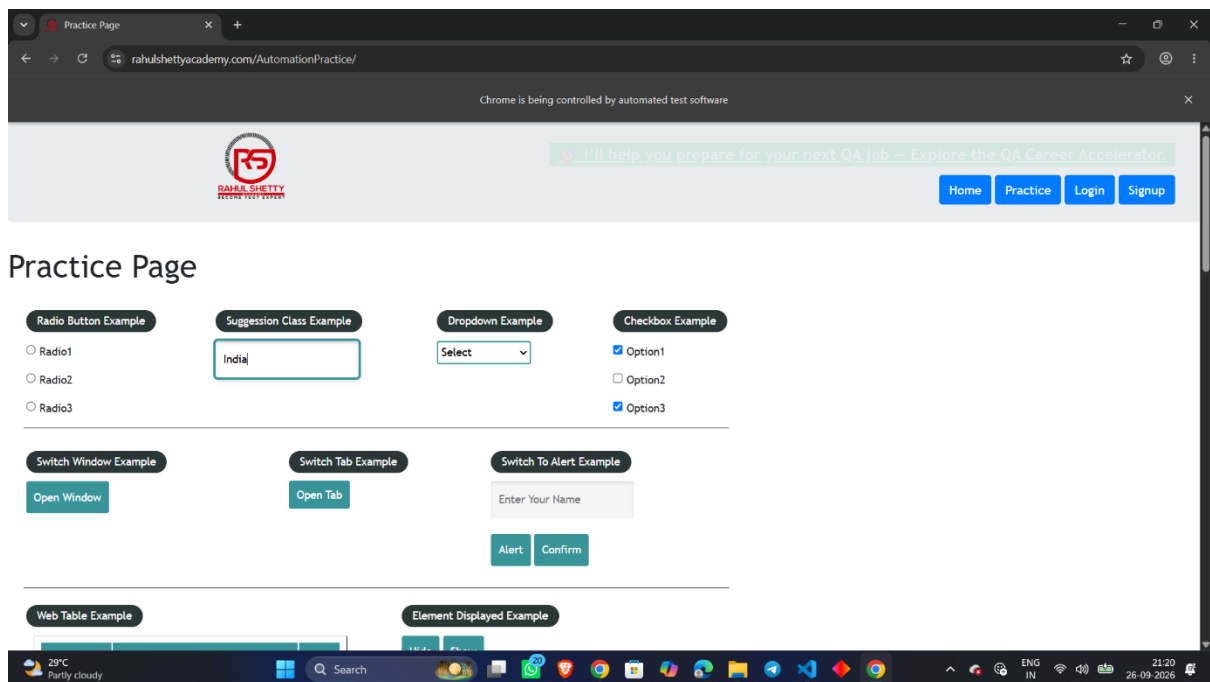
Screenshot 2: Multiple Checkboxes Selected

Description: The browser displays the Automation Practice website after Option 1 and Option 3 have been selected successfully. This screenshot demonstrates the checkbox selection performed by the Selenium automation script.



Screenshot 3: Autocomplete Selection

Description: The autocomplete field contains the selected country, India. The Selenium script entered `Ind`, displayed the available suggestions, and selected India from the dropdown.

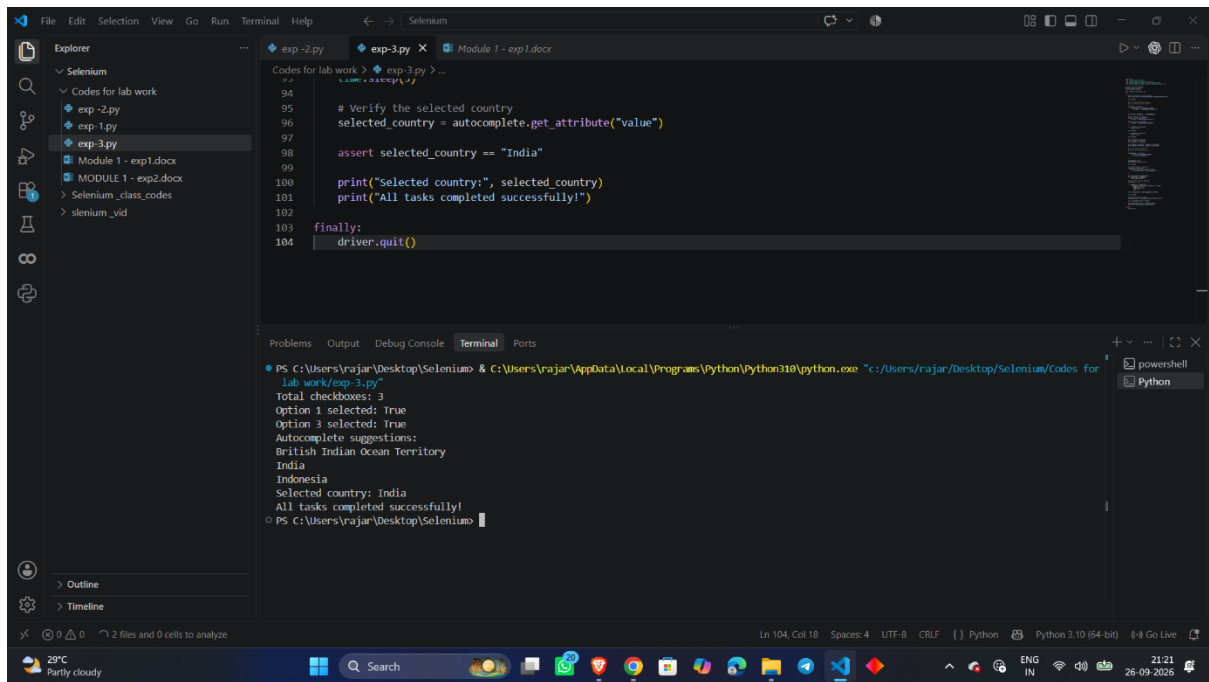


Screenshot 4: Successful Execution and Console Output

Description: The VS Code terminal displays the execution results. The output confirms that both checkboxes were selected, autocomplete suggestions were displayed, and India was successfully selected. The final success message indicates that all tasks were completed.

Console Output:

```
Total checkboxes: 3
Option 1 selected: True
Option 3 selected: True
Autocomplete suggestions:
British Indian Ocean Territory
India
Indonesia
Selected country: India
All tasks completed successfully!
```



7. Result, Observation and Conclusion

Result

The experiment was successfully executed using Selenium WebDriver with Python on the Rahul Shetty Academy Automation Practice website.

The automation script successfully performed the following tasks:

1. Opened the Automation Practice website.
2. Located and counted the available checkboxes.
3. Selected Option 1 and Option 3.
4. Verified that both checkboxes were selected.
5. Entered `Ind` into the autocomplete field.
6. Retrieved the dynamically displayed suggestions.
7. Selected India from the suggestions.
8. Verified that the selected country was India.
9. Displayed the successful execution message in the terminal.

Observation

- Selenium WebDriver successfully interacted with the checkbox elements.
- The `is_selected()` method verified the selection state of the checkboxes.
- The autocomplete field displayed suggestions after text was entered.
- Explicit waits helped the script wait for the autocomplete suggestions to become visible.
- Assertions confirmed that the checkboxes and selected country met the expected conditions.

- The script completed successfully and displayed the expected console output.

Conclusion

This experiment demonstrated how Selenium WebDriver can be used to automate multiple checkboxes and dynamic autocomplete dropdowns on a web page.

The experiment provided practical experience with element identification, checkbox selection, explicit waits, dynamic suggestion handling, and assertion-based verification.

The successful execution of the script demonstrated the use of Selenium WebDriver for automating interactive web elements and validating their behavior.